

WEEK 4 AND 5

CloudSim

Session 4

Installation of CloudSim Simulator

CloudSim is a framework for simulating cloud computing environments and evaluating different cloud strategies.

It is implemented in Java and requires Java and Eclipse IDE for execution.

⇒ Prerequisites:-

Before installing CloudSim, the system must have:

- 1) Java Development Kit (JDK) 8 or higher
- 2) Eclipse IDE or any Java-supporting IDE
- 3) CloudSim source code
- 4) Apache Commons Math library

Step 1:- Install Java Development Kit

CloudSim is written in Java, so JDK must be installed.

- Download JDK from the official Oracle website
- Install using installation wizard
- Set environment variable Path to JDK bin directory
- Verify installation using command:

```
java -version
```

Step 2:- Install Eclipse IDE

CloudSim requires an IDE to manage & run Java projects.

- Download Eclipse IDE
- Install and launch Eclipse
- Create a new Java workspace.

Step 3:- Download CloudSim

Cloudsim is open source and can be downloaded from GitHub

- Download CloudSim ZIP file
- Extract it to a directory such as C:\Cloudsim

Step 4 :- Identify Modules Folder

Inside extracted folder open modules directory which contains :-

- cloudsim (core library)
- cloudsim - examples (example programs)
- cloudsim - network (network module)

Copy the cloudsim folder

Step 5 :- Create Java Project in Eclipse

- Open Eclipse
- File → New → Java Project
- Enter project name
- Click Finish

CloudSim Simulation:

```
C:\Users\Admin>java --version
java 23.0.1 2024-10-15
Java(TM) SE Runtime Environment (build 23.0.1+11-39)
Java HotSpot(TM) 64-Bit Server VM (build 23.0.1+11-39, mixed mode, sharing)

C:\Users\Admin>|
```

Fig 1.1: JDK installation verification

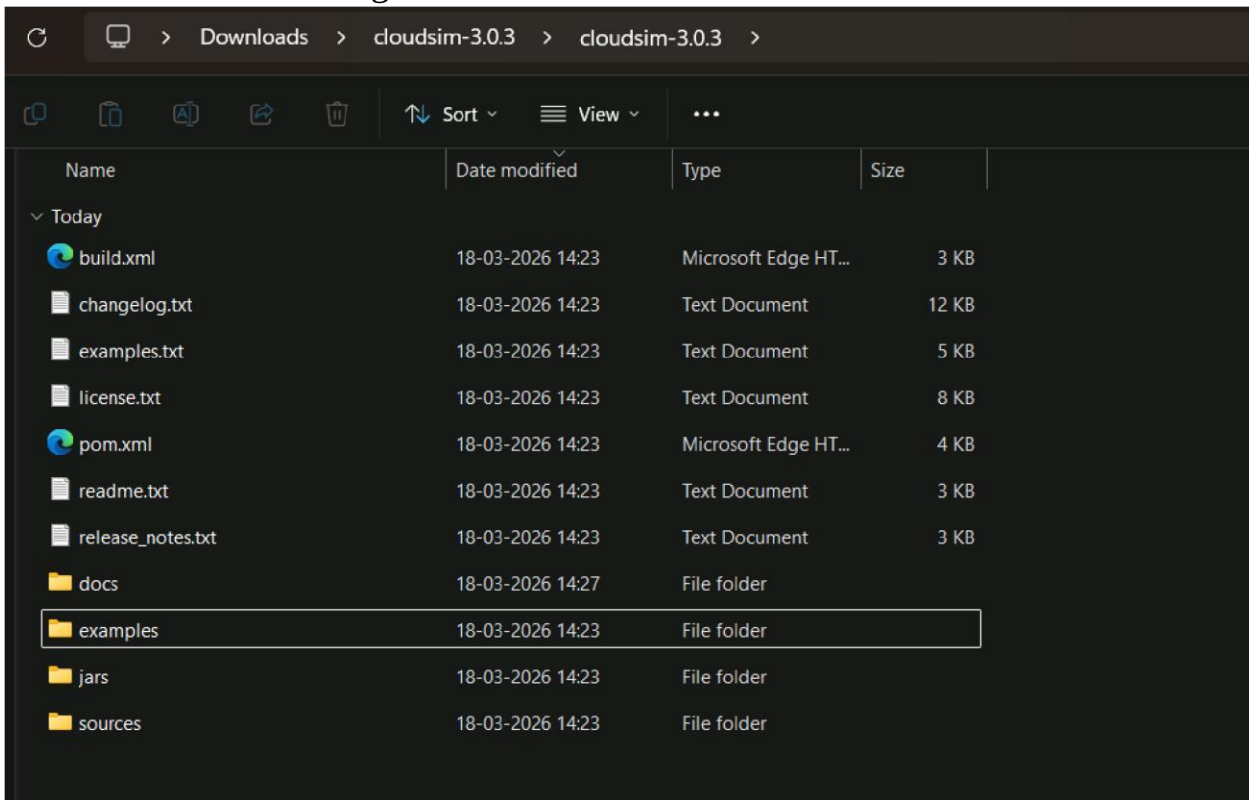


Fig 1.2: cloudsim zip folder extraction

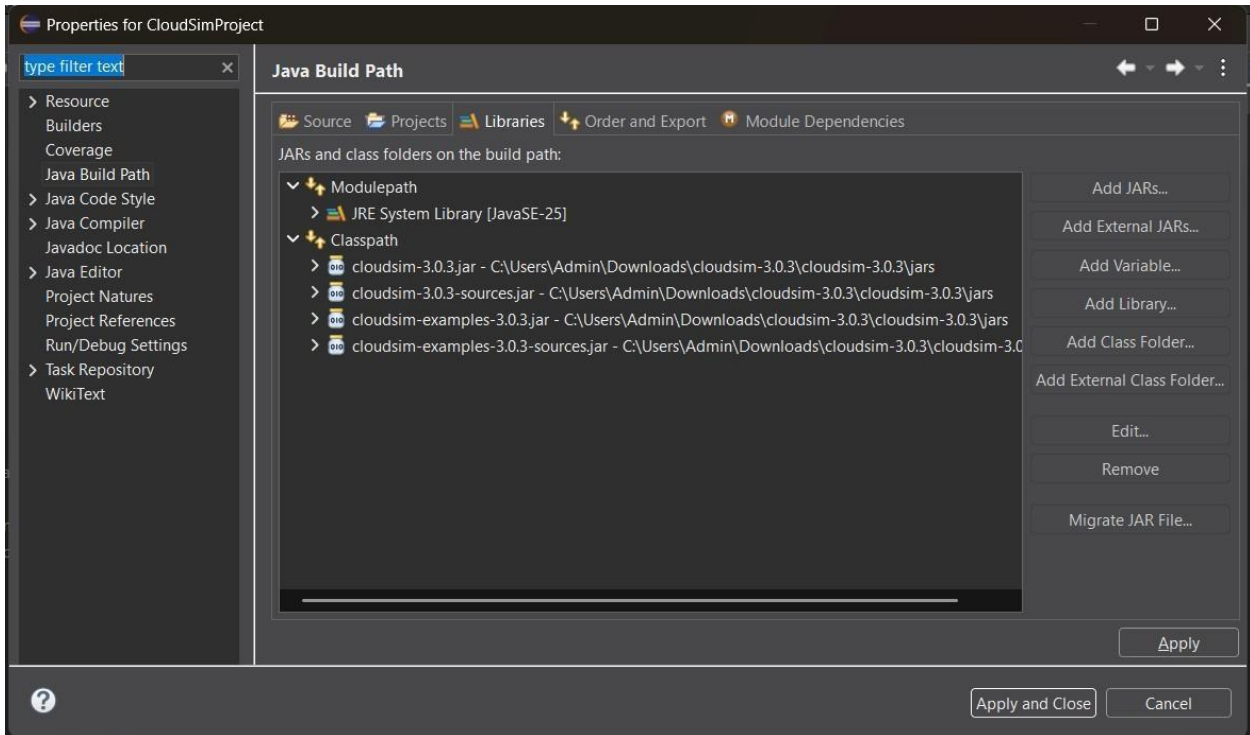


Fig 1.3: CloudSimProject properties

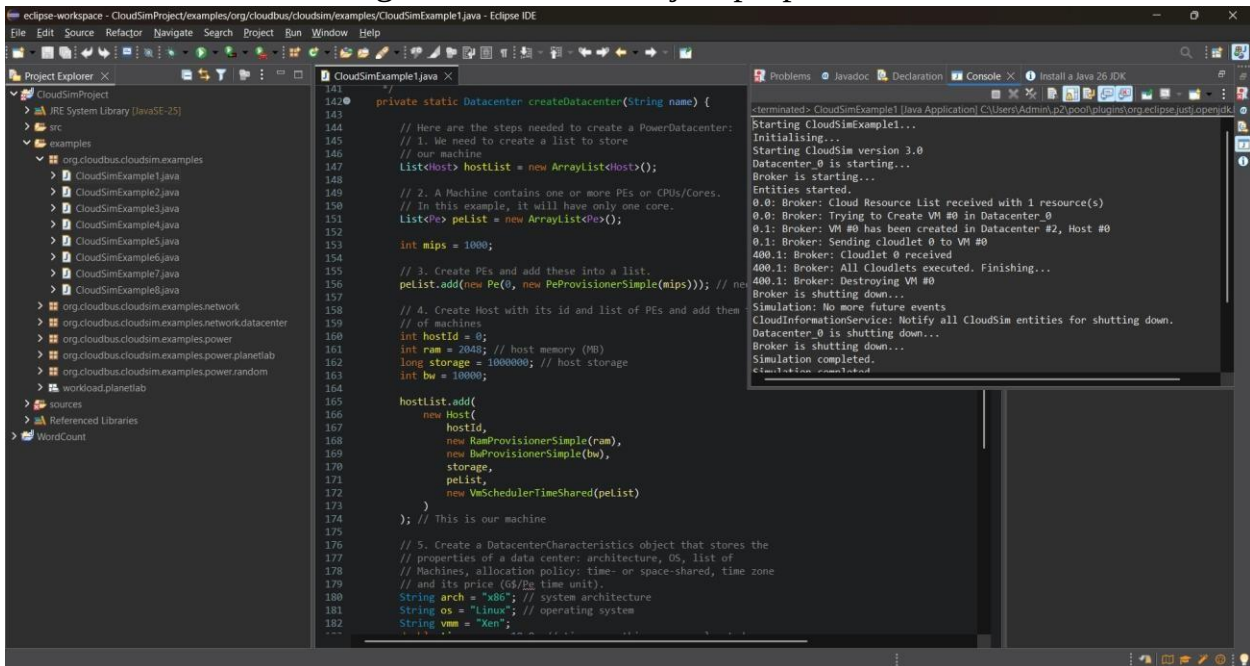


Fig 1.4: Running CloudSimProject - CloudSimExample1.java program

